

RESEARCH SURVEY

LLM-Based Intelligent Notification Composition

From Static Personalization to
Context-Aware Persuasive Messaging

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March 2026



Social Media



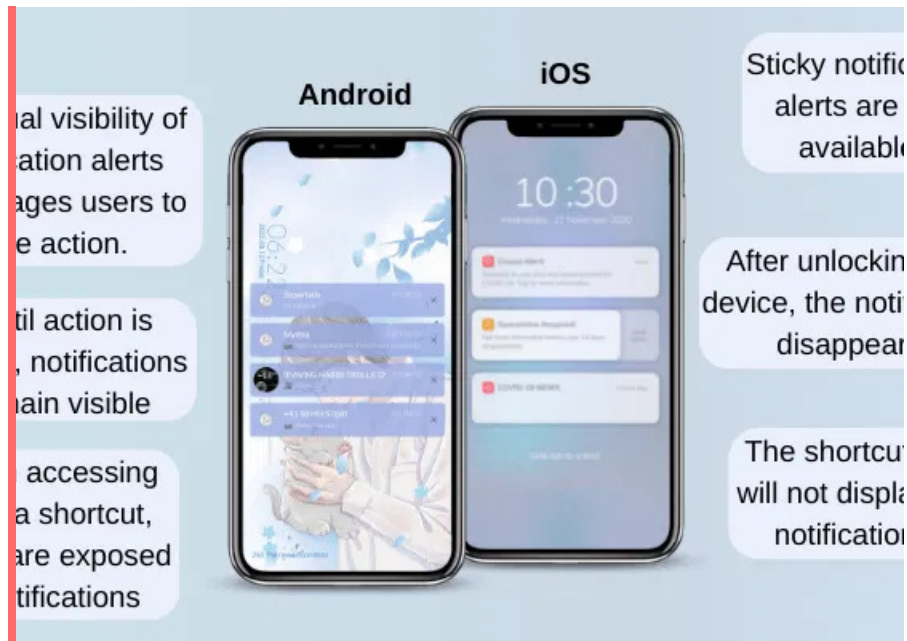
Food Delivery



E-Commerce

THE CORE PROBLEM

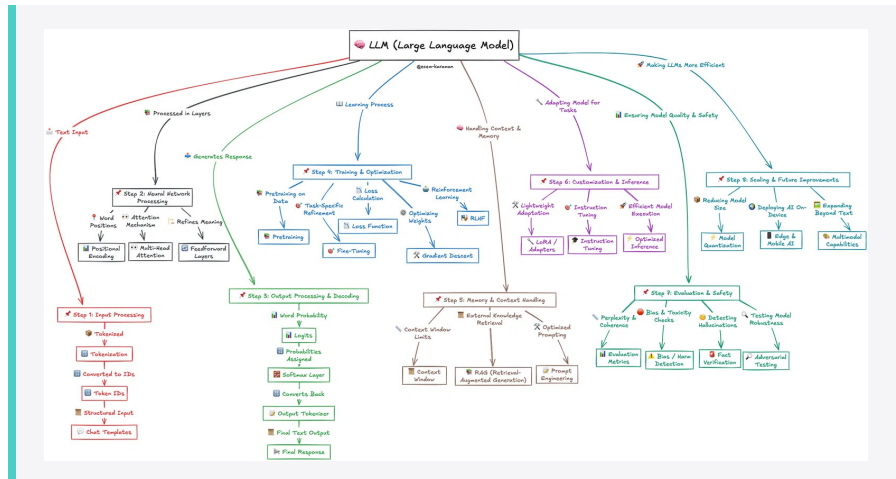
Notifications today are static, even when personalized



- Modern systems are highly effective at deciding **who** to notify, **when** to notify, and **what** to recommend.
- But they are weak at deciding **how** to communicate. Most rely on rigid, slot-filled templates (e.g., "Hi [Name], buy [Item]").
- This produces **relevance**, but not **intelligence**. The result is notification fatigue, eroded trust, and ignored messages.

The Gap: Transforming notifications from static, robotic reminders into adaptive, context-aware, and persuasive micro-recommendations.

The Shift to Intelligent Notification Composition



✗ Static Personalization (The Status Quo)

Systems select the right user, item, and time, but render the message through a fixed template. This produces **relevance without intelligence**.

Slot-Filling

Rigid Structure

Repetitive

The next evolution is not simply better personalization, but **intelligent notification composition**: using LLMs to transform static alerts into context-aware, persuasive messages.

✓ Context-Aware Messaging (The LLM Shift)

The message itself becomes adaptive. LLMs enrich the final act of communication, connecting user intent with **persuasive, action-oriented framing**.

Semantic Persuasion

Controlled Novelty

Real-Time Context

From Static Reminders to Persuasive Micro-Recommendations

WHO  User with Pizza Preference

WHEN  6:00 PM (Dinner Time)

WHAT  Pizza Category

Conventional System

"Order your favorite [Pizza] now."

- ✗ **Strategy:** Basic slot-filling into a fixed template.
- ✗ **Outcome:** Relevant targeting, but robotic and repetitive framing.
- ✗ **Limitation:** Identifies the user and item, but fails to persuade or engage intelligently.

Intelligent Composition (LLM)

*"It's the perfect time for dinner — your usual spot is nearby, and their **sweet-and-savory pineapple special** might be worth trying tonight."*

- ✓ **Strategy:** Contextual framing + controlled novelty.
- ✓ **Outcome:** Connects known preferences with a persuasive, action-oriented narrative.
- ✓ **Advantage:** Supports preference expansion by introducing adjacent options safely.

Review Methodology: 142 Sources → 28 Retained

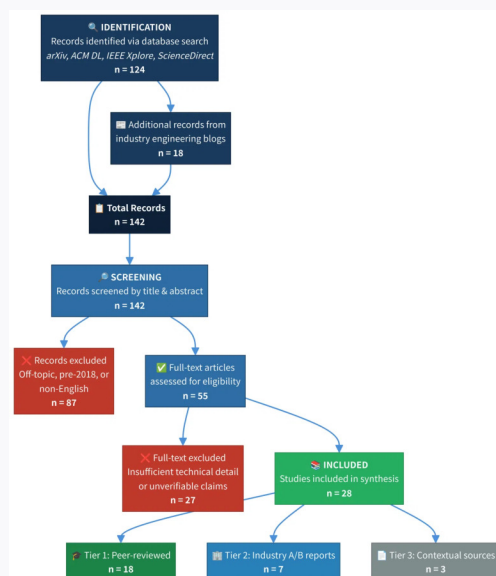
Systematic PRISMA review with 3-tier evidence classification ensures credible synthesis

PRISMA Screening Funnel

- 142** Sources (arXiv, ACM, IEEE, ScienceDirect, blogs)
- 87** Excluded: off-topic, duplicates, pre-2018
- 55** Assessed for eligibility
- 27** Excluded: low technical depth
- 28** Retained as primary sources

Evidence Tiers

- [T1] Tier 1** Peer-reviewed papers — full experiments & ablations
- [T2] Tier 2** Official engineering blogs — A/B results
- [T3] Tier 3** Vendor documentation — framing only



Search: Jan 2018 – Mar 2026

Keywords: LLM, RAG, PEFT, LoRA, CTR

The Critical Insight: LLMs Are One Stage in a 5-Stage Stack

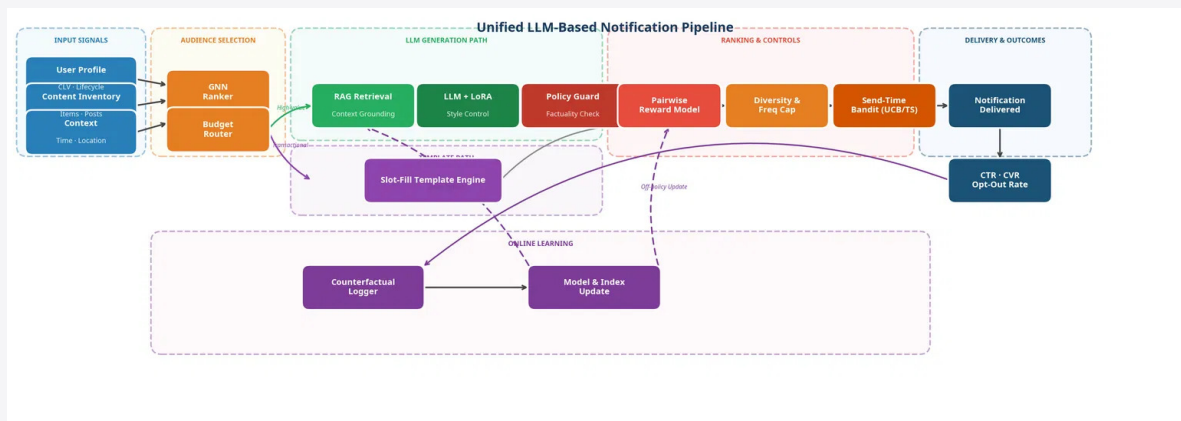
Observed engagement gains are often misattributed to text generation — the real driver is the full pipeline

COMPONENT	WHAT IT DECIDES	TECHNOLOGY
Audience Selection	WHO to notify	CLV models, churn prediction
Content Ranking	WHAT to notify about	Collaborative filtering, GNNs
Send-Time Optimization	WHEN to send	Contextual bandits
Message Generation	HOW to phrase it	LLMs, RAG, PEFT
Candidate Ranking	WHICH candidate to send	Pairwise reward models

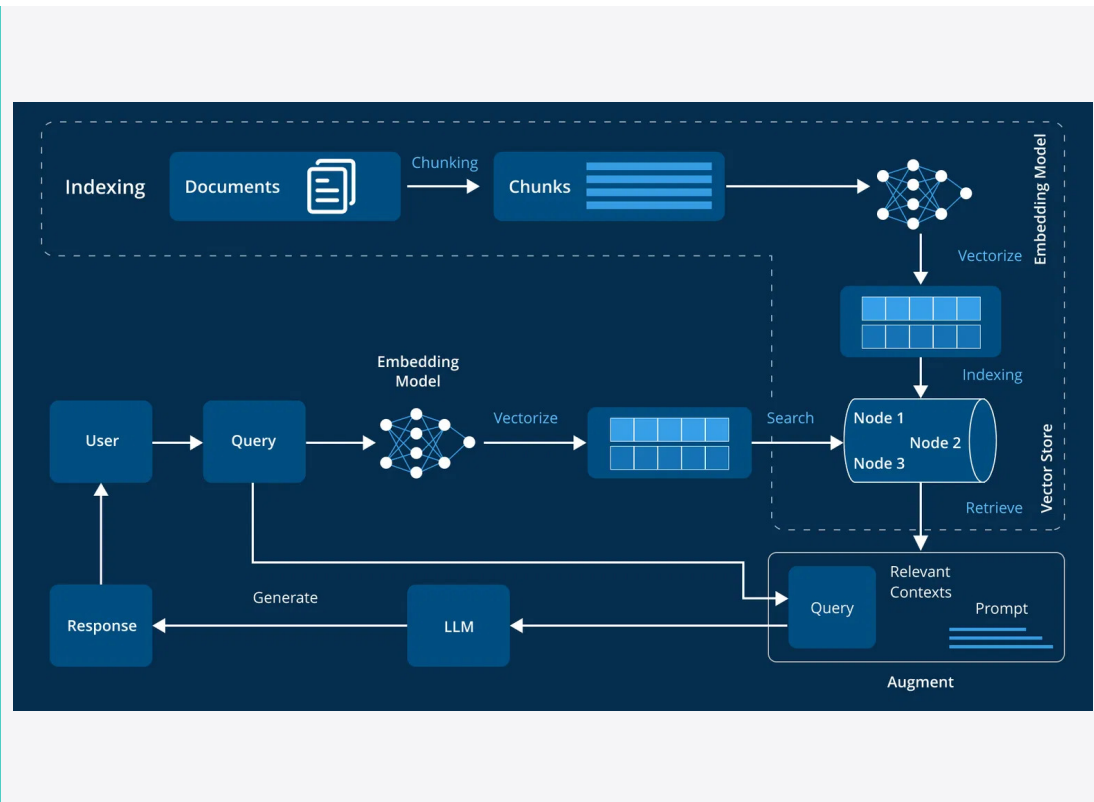


KEY WARNING

A system that generates brilliant copy for the wrong user, about the wrong item, at the wrong time will still fail. Conversely, a well-ranked, well-timed notification may perform strongly with simple language.



RAG Grounds LLM Output in Real Context



⚙️ How RAG works in notifications

- ✓ At inference time, RAG injects **user profile**, **item metadata**, **behavioral context**, and **campaign constraints** into the prompt.
- ✓ The LLM generates copy anchored to retrieved context, not just parametric memory.

"Rainy lunch? Your favourite Thai spot has 20% off until 2pm — Pad Thai delivered in 25 min"

⚠️ Key Tradeoffs

Retrieval granularity: Small chunks → higher precision, less context; large chunks → more context, more noise and latency.

Faithfulness hallucination: Even when correct context is retrieved, the model may blend it with parametric memory → unsupported claims.

Implication: RAG is a grounding mechanism, not a complete factuality solution — downstream verification is always required.

LoRA adapters enable hot-swappable tone control across campaigns with <1% parameter overhead

! The Problem

Full fine-tuning is expensive and operationally inflexible. Different campaigns, brands, and user segments require distinct tones, making monolithic models impractical for diverse notification streams.

⚙️ LoRA Mechanism

- ❄️ Freeze pre-trained model weights (W_0)
- + Inject trainable low-rank decomposition matrices

$$W = W_0 + BA$$

- 🔧 Train only B and A — typically **<1%** of total parameters
- ↔️ Adapters can be **swapped or merged** at deployment with zero inference latency overhead

📦 Applications in Notifications

- ✓ **Contextual Tone Shifting:** Urgency tone for cart abandonment versus friendly, casual tone for social re-engagement.
- ✓ **Brand Voice Consistency:** Maintaining strict stylistic guidelines across millions of generated messages for different sub-brands.
- ✓ **Rapid Experimentation:** Hot-swappable adapters allow for seamless A/B testing of different stylistic priors without deploying new base models.

⚠️ Key Risk

Over-optimization toward a narrow tone (e.g., maximizing CTR) can push the system toward formulaic, repetitive, or manipulative language patterns, ultimately causing notification fatigue.

Pairwise Reward Modeling Outperforms Pointwise CTR Prediction

✗ Pointwise Prediction (The Old Way)

Attempts to predict the absolute probability of a click for a single message.

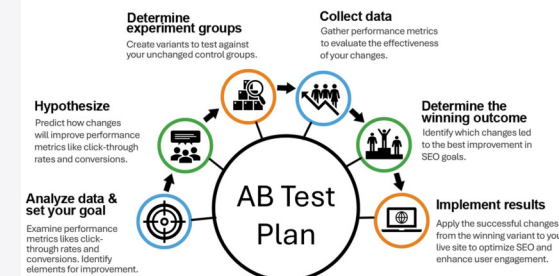
Why it fails: Highly vulnerable to confounding variables. A message might get a high CTR simply because it was sent at 12:00 PM to a highly engaged user, not because the text itself was good.

VERSUS

✓ Pairwise Modeling (The New Standard)

Takes two candidate messages generated for the **exact same user context** and predicts which one is more likely to win.

Why it wins: By holding the user, item, and time constant, it isolates the causal effect of the *linguistic variation*.



⚠ Failure Mode: Reward Hacking

Reward models are susceptible to **Entity Salience Bias**. If historical data shows that mentioning famous brands increases CTR, the reward model will score any message mentioning that brand highly, even if the brand is irrelevant to the current user's context.

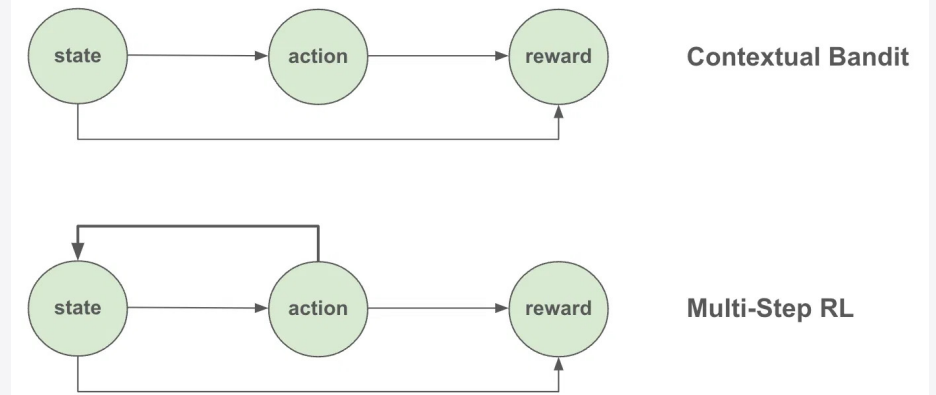
Solution: Factuality and policy guardrails must be applied **before** the reward model scores candidates.

Contextual Bandits Solve Send-Time Optimization

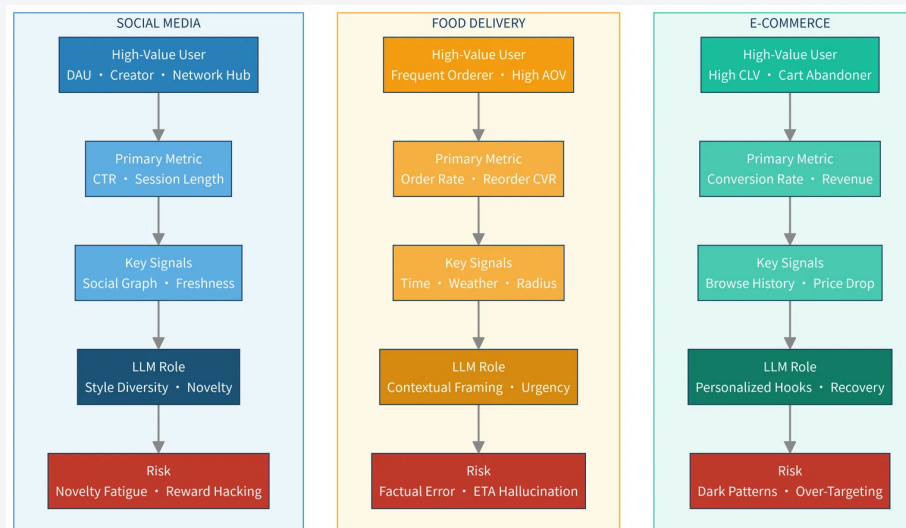
🕒 Beyond Static Timezones

- ✗ **The Failure of Heuristics:** Static rules like "send at 9:00 AM local time" fail to capture individual user routines, shift-work schedules, or weekend variations.
- ✂ **Thompson Sampling:** A probabilistic approach that balances **Exploitation** (sending at historically high-engagement times) with **Exploration** (testing new times to discover changing habits).
- ❄ **The Cold-Start Problem:** For new users with no history, bandits leverage contextual features (e.g., acquisition channel, device type) to infer optimal initial send times.

Even the most perfectly composed LLM message will fail if delivered when the user is asleep or driving. Send-Time Optimization (STO) is a critical prerequisite for message generation.



Three Domains, Three Definitions of High Value



Social Media

Goal: Habit formation & retention. **High Value:** Marginal users at churn risk.

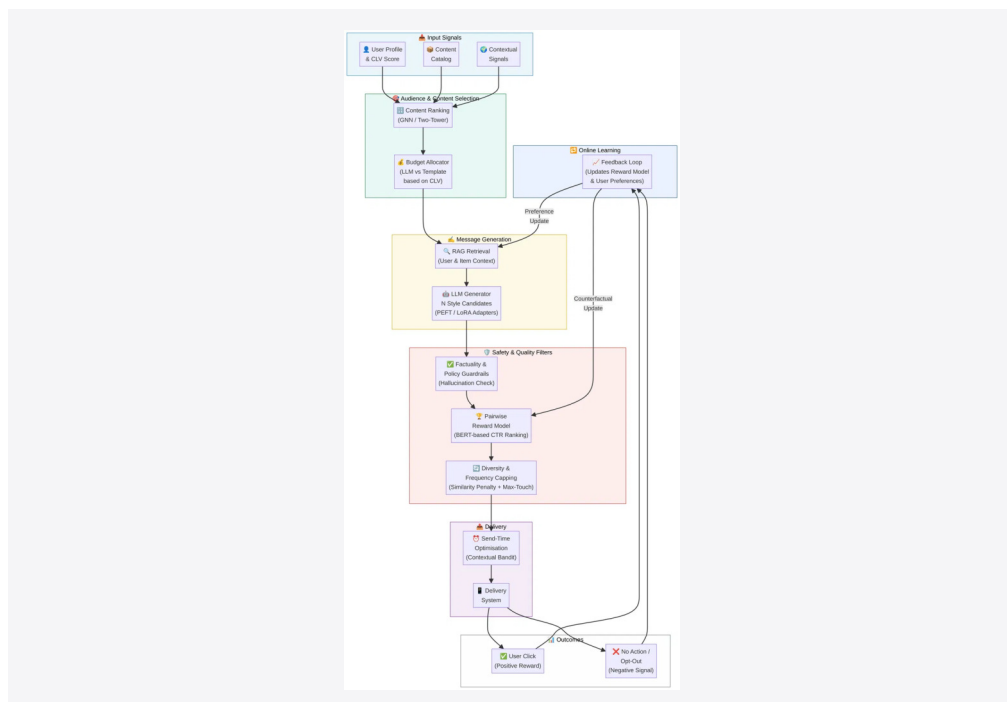
Food Delivery

Goal: Immediate conversion. **High Value:** Routine, time-sensitive users.

E-Commerce

Goal: High-AOV purchases & recovery. **High Value:** High-intent cart abandoners.

Case Study: A Food Delivery Re-Engagement Notification



1 AUDIENCE SELECTION

Identifies User U123: High CLV, dormant for 7 days.

2 CONTENT RANKING

GNN predicts highest affinity for **Thai Food**.

3 CONTEXT RETRIEVAL

RAG fetches: "Raining locally", "20% off promo active".

4 SEND-TIME OPT.

Bandit selects **11:45 AM** based on past lunch orders.

5 GENERATION

LLM + LoRA generates 5 candidate messages in a casual tone.

6 SAFETY FILTER

Regex checks drop candidates promising "free delivery" (false).

7 REWARD MODEL

Scores remaining candidates; selects the highest predicted CTR.

8 DELIVERY & LOG

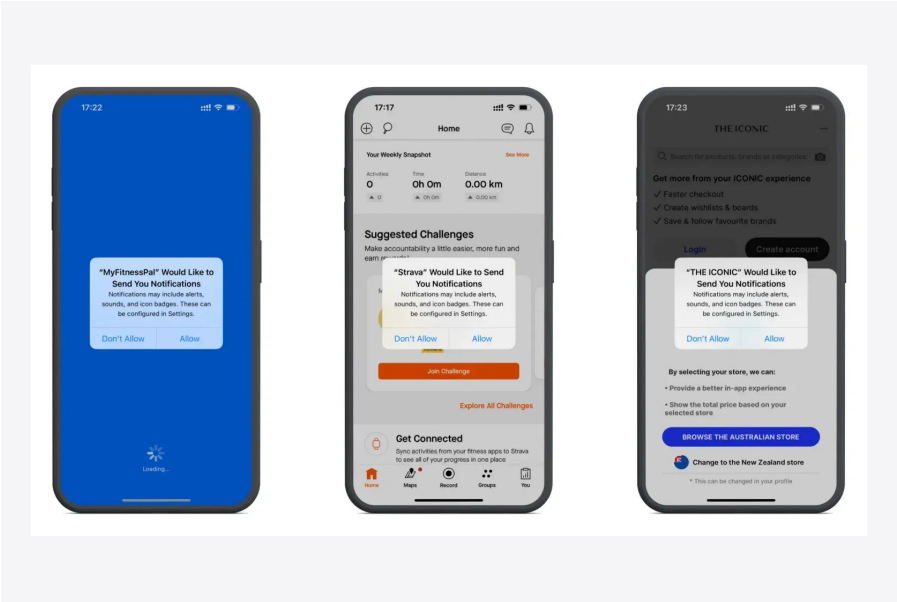
Message sent. Outcome logged for online learning update.

"Rainy lunch? Your favourite Thai spot has 20% off until 2pm — Pad Thai delivered in 25 min 🍜"

When LLMs Are NOT the Binding Constraint

A decision framework for when LLM generation adds value versus using templates

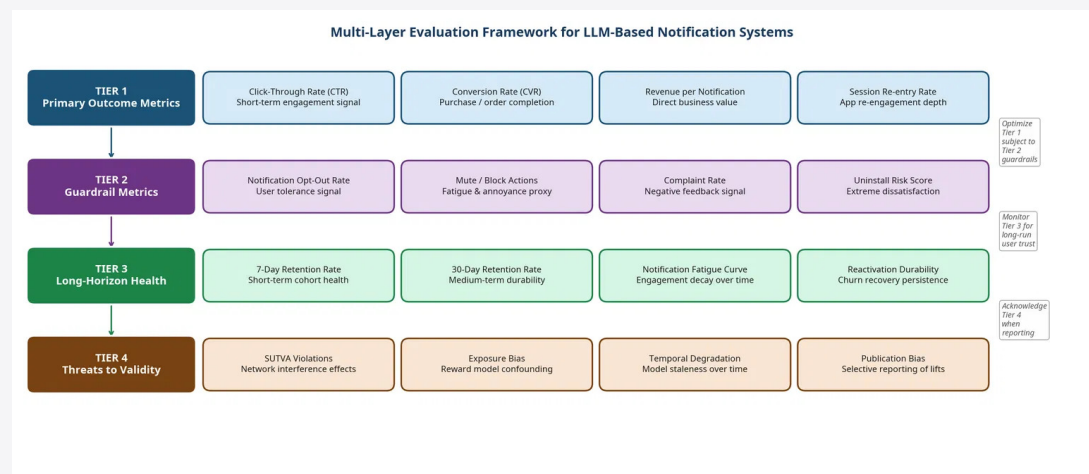
NOTIFICATION TYPE	LINGUISTIC VARIANCE	FACTUAL RISK	RECOMMENDED STRATEGY
Transactional (e.g., Order Shipped)	LOW	HIGH	Strict Templates
Time-Critical (e.g., Driver Arriving)	LOW	HIGH	Strict Templates
Price Drop / Inventory Alert	MEDIUM	HIGH	Template + Slot Filling
Social Re-engagement	HIGH	LOW	LLM Generation
Content Discovery (e.g., New Shows)	HIGH	LOW	LLM Generation
Abandoned Cart Recovery	HIGH	MEDIUM	LLM Generation + Guardrails



💡 The Budget-Aware Router

LLM inference is expensive and introduces latency. A mature architecture uses a **Budget-Aware Router** to direct high-CLV users or high-variance campaigns to the LLM, while routing transactional messages to deterministic templates.

The Unified Architectural Framework



! Key Failure Mode Tradeoffs

🛡️ Guardrail Throttling

Strict factuality and policy filters prevent brand damage but can **block too many valid messages**, forcing the system to fall back to generic templates and reducing overall campaign ROI.

🎯 Exposure Bias

The reward model only learns from messages that were actually sent. Over time, it becomes **blind to novel linguistic patterns**, leading to a feedback loop that reinforces repetitive, formulaic copy.

⌚ Latency Overhead

LLM generation adds significant latency. If generation takes too long, the message may **miss the optimal send-time window** identified by the contextual bandit, negating the value of the personalized copy.

Evaluation Must Go Far Beyond Short-Term CTR

Domain Generalizability Matrix			
Dimension	Social Media	Food Delivery	E-Commerce
"High-Value" User Definition	High-DAU user with strong social graph; influencer or power user	Repeat orderer with high AOV; lapsed user with reactivation potential	High-CLV customer; abandoned-cart user with purchase intent
Primary Success Metric	Engagement Rate (likes, shares, comments) Session re-entry	Order Conversion Rate GMV per notification Reactivation rate	Click-to-Purchase CVR Revenue per notification Cart recovery rate
Key Contextual Signals	Social activity feed Friend/follower events Content freshness	Time of day (meal windows) Order history & recency Location & weather	Browse & cart history Price drop / stock alerts Seasonal promotions
LLM Role in Pipeline	Personalise social recap & trending content summaries	Generate contextual urgency copy (e.g., lunch offers)	Produce persuasive product descriptions & recovery messages
Persuasion Tolerance	Moderate — users expect social updates, not hard selling	High during meal windows; low outside peak hours	High for deal/discount frames; low for generic promotions
Dominant Failure Mode	Notification fatigue from over-frequency; tone mismatch	False urgency (dark pattern risk); stale context	Hallucinated prices or stock levels; reward hacking
Cross-Domain Transferability	GNM audience selection transfers; LLM copy style does not	STO bandit transfers; urgency framing is domain-specific	RAG grounding transfers; CLV segmentation transfers broadly

Shaded left bars indicate row dimension category. Green cells = transferable across domains, white cells = domain-specific.

🎯 Tier 1: Primary Metrics

Short-term engagement: Click-Through Rate (CTR), Conversion Rate (CVR), and immediate revenue. While easy to measure, optimizing solely for these leads to clickbait and reward hacking.

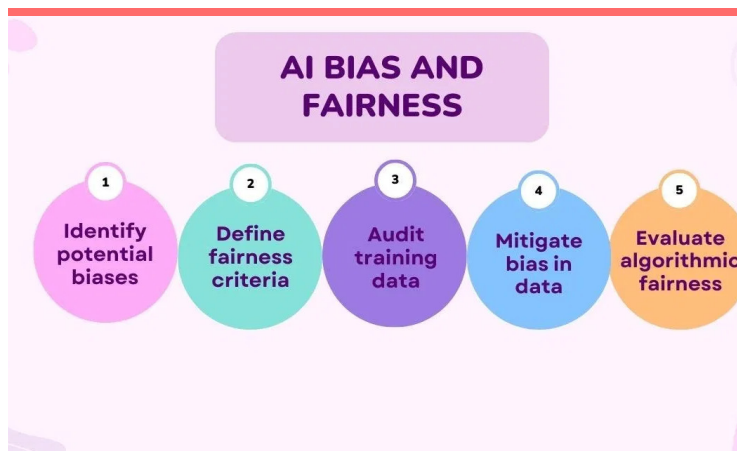
🛡️ Tier 2: Guardrail Metrics

User autonomy & safety: Opt-out rates, notification permission revocations, and negative feedback. These must be strictly bounded during A/B testing.

📈 Tier 3: Long-Horizon

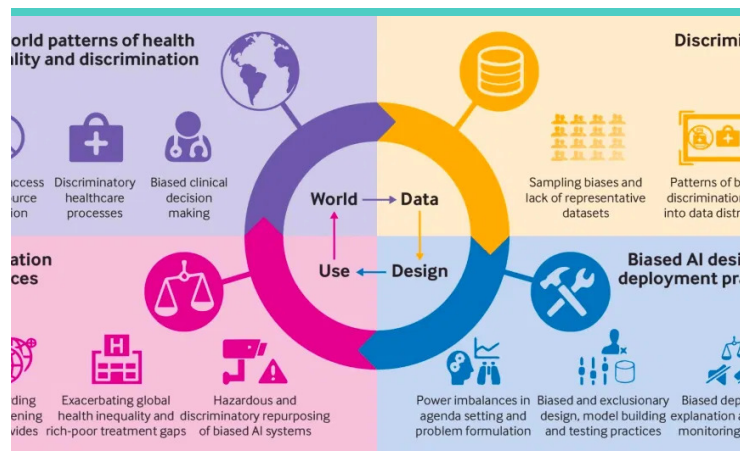
Ecosystem health: Customer Lifetime Value (CLV), 90-day retention, and platform uninstall rates. These capture the delayed effects of notification fatigue and trust degradation.

Ethical Boundaries: Personalization vs. Manipulation



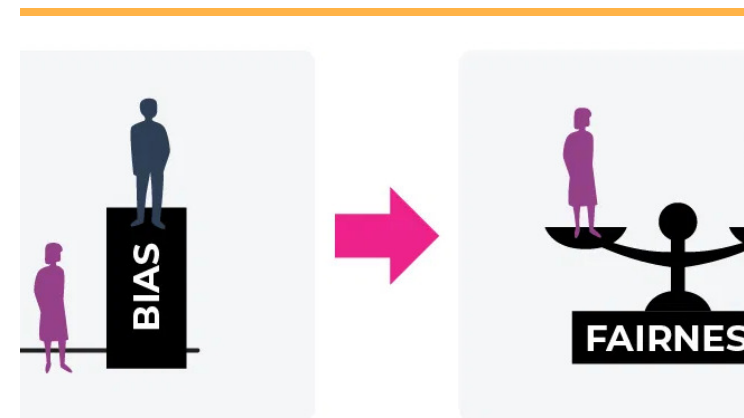
Persuasive Asymmetry

Platforms possess vast informational advantages over users. LLMs can weaponize this asymmetry by generating hyper-personalized "**dark patterns**" (e.g., false urgency or scarcity) that cross the line from helpful nudges into coercive manipulation.



Differential Susceptibility

Not all users respond equally to persuasive text. Reward models optimized purely for CTR may inadvertently learn to target **vulnerable demographics** with aggressive messaging, raising severe fairness and algorithmic bias concerns.



Regulatory Compliance

Under **GDPR Article 22** and **DSA Article 40**, automated systems with significant user impact require strict auditability. Black-box LLM generation complicates compliance, necessitating transparent guardrails and human-in-the-loop oversight.

Quantitative Evidence: Lifts Range from +1% to +14%

**+1.0% to
+14.5%**

Reported CTR Improvement

💡 The Baseline Effect

The massive variance in reported lifts is primarily driven by the sophistication of the **baseline system** being replaced, not just the capability of the LLM itself.

Low Baseline (Rule-Based / Static)

+10% to +14%

Scenario: Replacing static, one-size-fits-all templates with personalized LLM generation.

Result: Massive gains. LLMs easily outperform generic copy by introducing basic relevance and variety.

Medium Baseline (Basic ML / Slot-Filling)

+4% to +7%

Scenario: Replacing basic slot-filling (e.g., "Hi [Name], buy [Item]") with context-aware generation.

Result: Moderate gains. LLMs provide better fluency, tone adaptation, and natural integration of multiple context variables.

High Baseline (Mature ML / Bandits)

+1% to +2%

Scenario: Adding LLM generation to a pipeline that already has perfect audience targeting and send-time optimization.

Result: Marginal gains. When the right user gets the right item at the exact right time, the specific phrasing matters much less.

Threats to Validity and Future Research Directions

Threats to Validity



Publication Bias

Positive results are heavily overrepresented in industry engineering blogs, while failed experiments and negative A/B tests are rarely published.



Temporal Degradation

Initial CTR lifts often stem from a "novelty effect" as users react to new linguistic patterns. These gains typically degrade over time as users habituate.



Domain Representation

The current literature is heavily skewed towards e-commerce and social media, limiting generalizability to high-stakes domains like healthcare or finance.



Evaluation Confounding

Reported performance gains often reflect the combined impact of the entire pipeline (e.g., better targeting + LLM), making it difficult to isolate the LLM's specific contribution.



Future Research Priorities



Multimodal Generation

Moving beyond text to jointly generate personalized copy, images, and rich media layouts tailored to individual user preferences.



Causal Inference Methods

Developing robust frameworks to bridge the gap between offline reward model evaluation and online A/B test performance.



Federated Learning

Training personalization models directly on edge devices to ensure privacy compliance while maintaining high-quality targeting.



Direct Preference Optimization (DPO)

Exploring DPO as a more stable and efficient alternative to traditional Reinforcement Learning from Human Feedback (RLHF) in notification systems.



Cross-Domain Transfer

Investigating how stylistic priors and user behavioral models learned in one domain (e.g., social media) can transfer to another (e.g., e-commerce).

Five Key Takeaways

01 LLMs are not a silver bullet.

They are merely one component in a 5-stage pipeline. Audience selection and send-time optimization are equally critical.

02 Pairwise reward modeling is essential.

Pointwise CTR prediction fails due to confounding variables. Pairwise models isolate the causal effect of linguistic variation.

03 RAG and LoRA drive personalization.

RAG grounds generation in real-time context, while LoRA adapters enable hot-swappable tone control with minimal overhead.

04 Evaluation must look beyond short-term CTR.

Optimizing solely for clicks leads to reward hacking. Guardrail metrics (opt-outs) and long-horizon metrics (CLV) are mandatory.

05 Ethical guardrails are operational necessities.

Factuality filters and frequency capping must be implemented to prevent manipulation, brand damage, and notification fatigue.

CALL TO ACTION

The research community must shift focus from isolated text generation to **end-to-end pipeline optimization**.

Future work should prioritize causal inference in reward modeling, longitudinal studies on notification fatigue, and robust frameworks for cross-domain transferability.